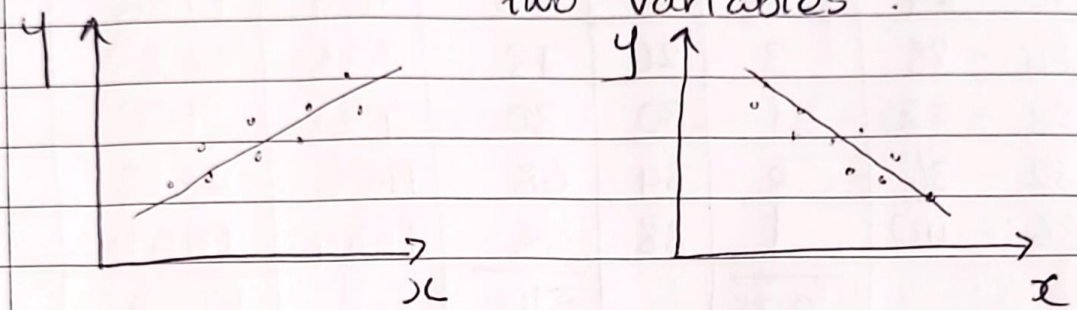
Jan 2019

Q3 Statistics & Probability.

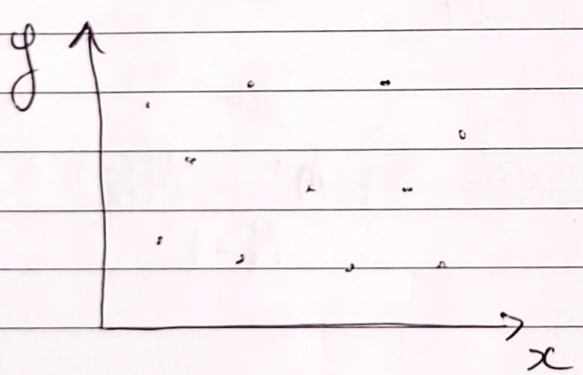
a) Correlation – statistical measure
– measures extent to which two variables are related (linearly)
– describes the relationship between two variables.



positive correlation
i.e. y increases as x increases

negative correlation
i.e. y decreases as x increases.

Figure 3.1 → negative correlation.



no correlation.

ii) exception – data point that is distant from the main cluster of points or does not fit with the main trend.

in fig. 3.1. e.g team in 12th position scored more goals than the teams in position (7-11th)
the team in 14th position scored less goals than the teams that finished in lower positions...

b)

No. of Months	f_i Freq	M_i	$f_i M_i$	M_i^2	$f_i M_i^2$
0 - 4	4	2	8	4	16
4 - 8	8	6	12	36	72
8 - 12	3	10	30	100	300
12 - 16	3	14	42	196	588
16 - 20	4	18	72	324	1296
20 - 24	4	22	88	484	1936
24 - 28	3	26	78	676	2028
28 - 32	1	30	30	900	900
32 - 36	2	34	68	1156	2312
36 - 40	1	38	88	1444	1444
	27		516		10,892

i) Mean: $\bar{x} = \frac{\Sigma f_i M_i}{\Sigma f_i} = \frac{516}{27} = \boxed{19.1}$

(Freq, Midpoint)

ii) Standard deviation: $S^2 = \frac{\Sigma f_i M_i^2 - M(\bar{x})^2}{M-1}$

(Midpoint^2)

where $M = \Sigma f_i$

$$S^2 = \frac{10,892 - 27(19.1)^2}{27-1}$$

$$= \frac{10,892 - 9849.87}{26}$$

$$S^2 = 40.08$$

$$S = \sqrt{40.80} = \boxed{6.33}$$

c) lotto ⇒ order does not matter ${}^nC_r = \frac{n!}{(n-r)!\,r!}$

$n = 32$
$r = 4$

${}^{32}C_4 = \frac{32!}{(32-4)!\,4!}$

$= \boxed{35{,}960}$

d) normal distribution $a = \frac{X - \mu}{\sigma}$

$\mu = 28$
$\sigma = 8$

$P[10 < X < 40] = P[X > 10] - P[X > 40]$

$a = \frac{10 - 28}{8} = -2.25$

$P[X > 10] = P[Z > -2.25] = 1 - P[Z > 2.25]$
$= 1 - 0.0122$
$= 0.9878$

$P[X > 40]$ $a = \frac{40 - 28}{8} = 1.5$

$P[X > 40] = P[Z > 1.5] = 0.0668$

$P[10 < X < 40] = 0.9878 - 0.0668$

$= \boxed{0.921}$